

Yash Tarun

Adm. No. 20JE1114 +447900334978
✉ tarunyash01@gmail.com  linkedin.com/in/yash-tarun-522a7321a/  github.com/yashwork-byte



Education

Indian Institute of Technology(ISM) Dhanbad Dec 2020 - May 2025
Bachelor of Technology(Double Major) Computer Science and Engineering (GPA: 8.13 / 10.00) Dhanbad, Jharkhand
• **Relevant Coursework:** Data Structures and Algorithms (C++), Prob & Stat, Object Oriented Programming (C++), Operating Systems, Database Management Systems, Computer Network, Software Engineering, Deep Learning, Advanced Algorithms

Experience

Barclays UK | Python, SQL, BarclaysAI (Claude), Celonis, REST APIs Aug 2025 – Present
Technology Analyst Knutsford, United Kingdom
• Designed and implemented an **AI-driven incident intelligence layer** on top of NowAssist using BarclaysAI (Claude models), enhancing incident triage and resolution workflows across enterprise systems.
• Built pipelines to **auto-correct incident descriptions, recommend assignees, and suggest resolution strategies** by leveraging historical incident data from SPDW (via Celonis), improving triage accuracy by **30%** and reducing manual intervention.
• Architected a **context-aware recommendation system** to cluster similar incidents and trigger automated email notifications, reducing mean time to resolution (MTTR) by **20–25%** and improving assignment precision.

Barclays UK | Deep Neural Networks, Python, SQL June 2024 – August 2024
Technology Analyst Intern Knutsford, United Kingdom
• Developed **machine learning models** to automate **control and validation processes**, reducing manual review time by **20%** within the Group Technology Service Management (GTSM) team.
• Enhanced **model accuracy** from **75%** to **82%** primarily by switching from random forest models to Deep Neural Networks using rollback plans and improving dataset labeling efficiency by **15%**.
• Selected high-performing models after analyzing **performance metrics**, resulting in an increase in process efficiency.

Projects

CompCode – Codebase Understanding & Debugging System | Python, AST, Qdrant, OpenAI, Streamlit | Github
• Built an intelligent system to **analyze large codebases** using AST-based function extraction and semantic embeddings stored in a vector database.
• Developed a **hybrid retrieval pipeline** combining vector similarity with call graph expansion, incorporating upstream/downstream dependencies for improved contextual relevance, reducing irrelevant context retrieval by **30%** compared to pure semantic search.
• Executed a **multi-agent architecture** with routing, reasoning, and debugging agents, along with context compression and memory integration for multi-turn code understanding.

Distributed Task Processing System | Node.js, Redis Streams, PostgreSQL, WebSockets | Github
• Planned and created a **distributed task queue system** using Redis Streams and consumer groups, enabling parallel execution across multiple workers with horizontal scalability.
• Engineered **fault-tolerant retry and dead-letter queue (DLQ) pipelines** and architected a decoupled event-driven system using Redis for messaging and PostgreSQL as the source of truth.
• Built a **real-time observability layer using WebSockets** and addressed large scale systems challenges including **idempotency, duplicate processing, and consumer acknowledgment** under concurrent execution.

Sports Programming

Codeforces: Rating **1590 (Specialist)**
Codechef : Rating **1892 (4*)**
Leetcode: Rating **1686**

Technical Skills

Languages: C++, Java, C, Python
Technologies: Node.js, Spring Boot, REST APIs, Redis, WebSockets, Qdrant, PostgreSQL, MongoDB, Docker
Miscellaneous: Distributed Systems, Event-Driven Architecture, Microservices, LLM Systems, Retrieval-Augmented Generation (RAG), Vector Databases, Prompt Engineering, Git, Linux
Concepts: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, OOP, System Design

Achievements

CodeForces Round 970 Achieved Global Rank Under 100.
Codechef Round 144 Attained Global Rank 53.
Ranked 2nd in Hackfest IIT(ISM) 2022 organized by Atlassian.
Solved **more than 800+** problems across platforms.
Top **1%**ile in the JEE Mains and JEE Advanced 2020.

Social Engagements

Event Co-Head: Concetto'22, annual techno-management fest of IIT(ISM) Dhanbad.